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Review

A Review of Research on the Responses of Agricultural Households to Eco-Compensation in China

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Abstract

The responses of agricultural households are the central link in China's eco-compensation, which directly determines the efficiency and effectiveness of compensation. This article reviewed the connotation, influencing factors, and ultimate effectiveness of agricultural households' response to eco-compensation in China. The results indicated that agricultural households' response to eco-compensation mainly includes reducing production intensity, optimizing production methods, and changing livelihood types. On this basis, taking protected objects such as farmland, grassland, forests, and watersheds as examples, the specific connotation of the responses was explained. Subsequently, according to policy mechanisms, sustainable livelihood theory, and planned behavior theory, the factors that affect agricultural households' responses have been sorted out, forming a systematic factor system framework. In addition, focusing on the policy objectives of eco-compensation, the research progress on the economic and ecological effects of agricultural households' responses, and the final results were summarized. Finally, this article identifies four shortcomings in the current research, namely imbalanced research fields, incomplete contextual impact, nonspecific ecological effects, and immature improvement strategies. The future literature should strengthen research in key fields and areas, focus on the correlation between contexts and responses, integrate multiple disciplines to accurately evaluate ecological effects, and demonstrate the improvement mechanism of agricultural households' responses.

Keywords: eco-compensation; agricultural sustainability; agricultural household livelihood; agricultural ecological product value realization; payments for ecosystem services

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1. Introduction

Since the second half of the 20th century, China's economy has developed rapidly, but it also faces challenges in ecological sustainability [1]. In order to achieve higher yields, many agricultural systems have developed high-intensity and high-input farming practices, leading to ecological and environmental threats such as soil acidification [2], water pollution [3], massive emissions of greenhouse gases [4], and a decline in biodiversity [5]. In recent years, the Chinese government has placed great emphasis on ecological civilization

construction, striving to build a modern society where humans and nature coexist in harmony. To coordinate the relationship between protection and development and internalize the externalities of the ecological environment, eco-compensation has emerged [6].

China's eco-compensation is similar in connotation to the widely mentioned "Payments for Ecosystem Services (PES)" internationally [7–9], and it has been implemented in many ecosystems such as forests [10–12], grasslands [13,14], watersheds [15,16], and farmland [17]. Agriculture, as one of the industries with the closest interaction between humans and nature [18], has become an important area for China's industrial ecological transformation. The main mode of current agricultural eco-compensation practice in China is for the government to provide various supportive services and policies to agricultural households, stimulate them to make ecological responses, and indirectly restore the functions of the agricultural ecosystem [19,20]. The responses of agricultural households to eco-compensation are crucial. From a theoretical perspective, they are the key to interpreting the mechanism of agricultural eco-compensation and can serve as a policy reference. From a practical perspective, they are the central link in the compensation process, directly affecting the final efficiency and effectiveness.

With the deepening of academic research on agricultural households' responses in the background of eco-compensation, more and more studies have shown that the connotation of the responses is rich. Firstly, the ways in which agricultural households respond are diverse. The general compensation policy not only imposes restrictions on agricultural production but also provides guidance for transformation, which brings choices to agricultural households [21]. Secondly, each response is often accompanied by a complex decision-making process. This process is influenced by various factors such as land endowment [22], household income [23], and education level [24]. Finally, when the production methods of agricultural households change, a series of ecological and socio-economic impacts may arise [25]. Achieving the goal of compensation by guiding agricultural households to respond is a continuous task. However, most of the current literature only focuses on a certain aspect of research and lacks systematicity. In terms of the entire compensation process, systematic research has the following three advantages: Firstly, it helps to clarify the mechanisms of each link and form a more complete research system; secondly, it helps to comprehensively summarize various factors that affect the effects of compensation, providing scientific references for policy makers; thirdly, it can discover the shortcomings of current research from a broader perspective and provide ideas for future research. Therefore, this study aims to systematically review the relevant research on agricultural households' response under the background of China's eco-compensation policy, including its connotation and types, influencing factors, and ultimate effects.

2. Research Framework on Agricultural Households' Response to Eco-Compensation in China

2.1. Research Scope

The research object of this review was the responses of agricultural households in the context of eco-compensation in China. Broad agriculture includes four categories: crop cultivation, forestry, animal husbandry, and fishery. Therefore, in this review, agricultural households covered various planting households and breeding households involved in land management (Figure 1). Regarding the background of this review, according to China's *Regulations on Ecological Protection Compensation*, the scope of eco-compensation in China covers forests, grasslands, wetlands, deserts, oceans, watersheds, and farmlands, as well as other important ecological and environmental elements such as aquatic biological resources, terrestrial wildlife, and plant resources stipulated by laws, administrative regulations, and national stipulations. To match the production subject of agricultural households, this

study focused on eco-compensation mechanisms and policies related to agricultural development, namely forests, grasslands, cultivated lands, watersheds, and aquatic biological resources. The reasons are as follows: Firstly, the compensation in these fields has provided varying degrees of ecological transformation support for agricultural households' planting production [26] or breeding production [27,28]. This is consistent with the theme of this study. Secondly, these fields currently have complete policy mechanisms. Compared to other fields, their compensation theory and specific practices are more mature [29–32].

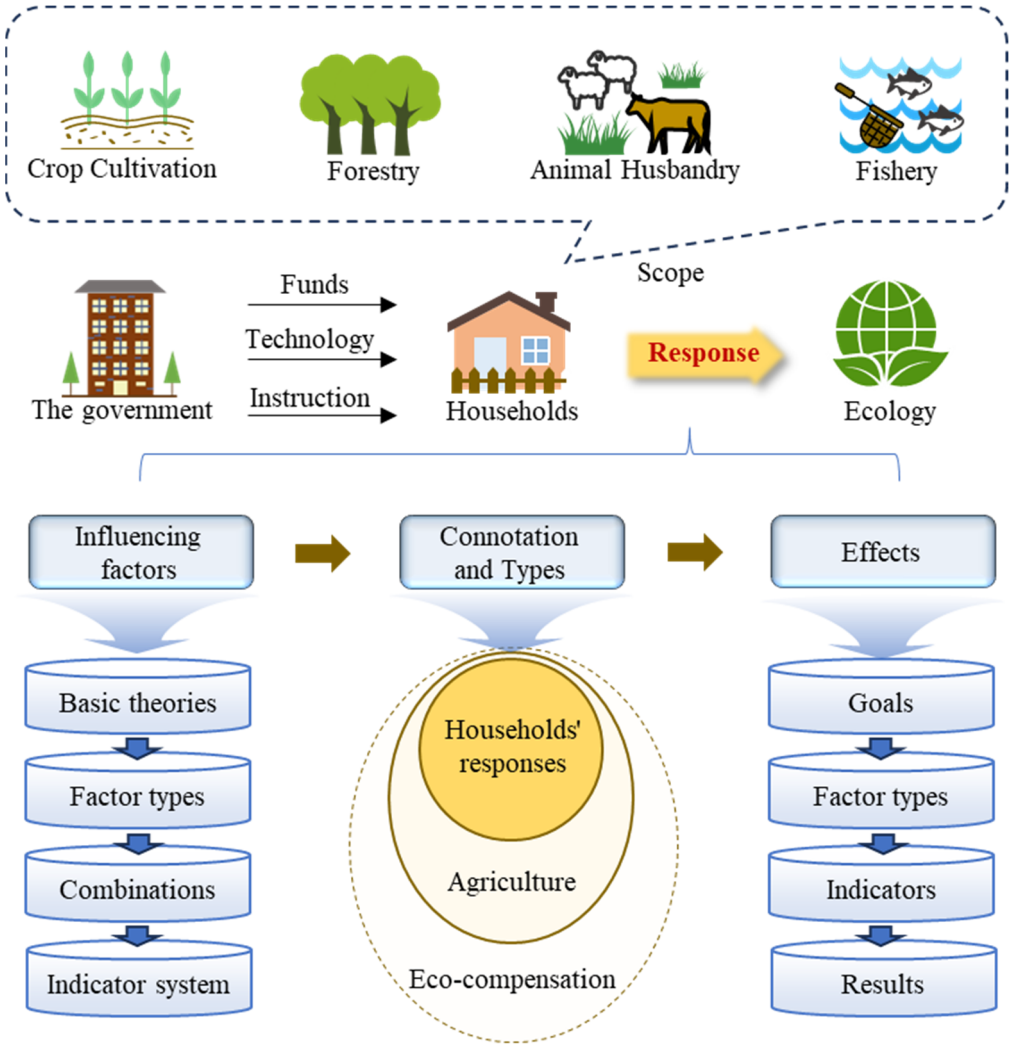

Figure 1. Schematic diagram of the research framework.

2.2. Core Contents

As mentioned earlier, the responses of agricultural households are the central link of eco-compensation. In order to systematically sort out the response of agricultural households, this study mainly focused on three aspects: Firstly, the most fundamental aspect is the connotation of the responses. Under the policy of eco-compensation, how do agricultural households respond? Subsequently, by reviewing a large amount of the literature, the "causes" and "results" of agricultural households' responses were analyzed. "Causes" are the influencing factors of agricultural households' response to compensation. This review systematically summarized the relevant basic theories, factor types, and analysis combinations, ultimately forming a factor indicator system. This provided a paradigm and reference for future research. "Results" are the diverse effects of agricultural households' responses. This is a direct evaluation of the eco-compensation policy. Research has shown

that eco-compensation is a public institutional arrangement that improves the ecological environment through socio-economic means, so the effects of compensation often have multiple aspects [33–35]. Therefore, this review summarizes the types of effects and practical results of agricultural households' response to eco-compensation. Finally, based on the conclusion, the shortcomings of the current research were pointed out, and suggestions for future research directions were proposed.

3. The Connotation and Types of Agricultural Households' Response to Eco-Compensation in China

Many countries and regions around the world have implemented practices of improving the ecological environment through economic incentives. PES directly corresponds to the concept of ecosystem services and is widely applied worldwide [36–38]. In practice, there are also some concepts that are similar to PES. One is the agri-environmental schemes (AES), which is regarded as a sub-type of PES, aiming to address Europe's environmental and climate goals [39–41]. Another is the eco-compensation mentioned in this study, which is widely applied in China. Although the concepts are similar, the most crucial feature of eco-compensation is that it is large-scale, government-led, and has already issued regulations [42,43].

In the general PES research literature, internationally, scholars often use "participation" to express the behavior of agricultural households (considered participants) supporting PES [44–46]. This is because PES emphasizes motivation more, and participants have a higher degree of voluntariness when choosing whether to support PES. However, in China, although many studies also use PES to refer to China's eco-compensation [47–49], it has certain characteristics in essence. The main differences between "response" in this study and "participation" in PES lie in two aspects. The first is the difference in responsibilities. In China, eco-compensation has always been promoted through large-scale institutional arrangements [50,51]. Some provisions in compensation policies are both incentives and obligations for agricultural households. Therefore, in this situation, using "response" to describe the support of Chinese agricultural households for eco-compensation could be more accurate. The second is the difference in connotation. "Participation" emphasizes more direct feedback to PES activities. The connotation of "response" in this study not only includes this kind of feedback but also encompasses other adaptive adjustments made by agricultural households in their production and livelihood in the context of eco-compensation.

This study divided the responses into two steps and three ways (Table 1). Firstly, the compensation policy clearly requires that when land resources such as farmlands, grasslands, forests, and watersheds are used for restrictive or protective purposes, agricultural households should take actions to comply with the policy requirements while receiving compensation support (funds, technology, and instruction). For example, in order to protect natural resources, fallow policies encourage agricultural households to stop long-term farming activities for a certain period of time [17]. The grassland ecological protection subsidy and reward policy encourages agricultural households to reduce grazing intensity in accordance with the requirements of grazing prohibition and grass-livestock balance [19]. The policy of grain for green encourages agricultural households to stop planting activities in specific areas [52]. The fishing ban policy encourages agricultural households to reduce fishing activities in key protected watersheds [53]. In the above situation, the most basic response of the compensated agricultural households is to reduce the corresponding production intensity according to policy regulations.

Subsequently, after reducing production intensity, agricultural households received compensation support and more working time. At this point, they can integrate these resources to make other responses. There are mainly two ways: Firstly, continue to engage in production activities related to the original according to the guidance of compensation (Figure 1). But this requires optimizing production methods according to policy requirements: that is, achieving protective use of land resources through ecological transformation. For example, the protective use of farmland can be manifested as optimizing agricultural inputs [54,55], promoting green manure planting [56], carrying out crop rotation [57], and comprehensive utilization of straw [58]. Protective grassland utilization can promote cross-regional grazing, popularize manual grass-cutting, and develop barn feeding [21]. Protective forest land use can develop agroforestry [59] and cultivate economic forests [60]. Protecting watersheds can promote the conversion of nearby paddy fields to dryland agriculture [61]. Secondly, agricultural households can change their livelihood types to expand new income channels. In this case, they may abandon their original agricultural production or continue to maintain low-intensity or ecological agricultural production.

In summary, by reviewing the research on agricultural households' responses under the background of eco-compensation in China, this study categorized agricultural households' responses into three types: reducing production intensity, optimizing production methods, and changing livelihood types. It should be emphasized that in actual production, these three types of responses are not mutually exclusive. Agricultural households may respond to eco-compensation in multiple ways simultaneously.

Table 1. Summary of types and measures of agricultural households' responses to eco-compensation in China.

Types	Protecting Objects	Specific Changes	References
Reducing production intensity	Farmlands	Fallow	Xie et al., 2022 [17]
	Grasslands	Reduce grazing	Li et al., 2022 [19]
	Forests	Stop farming	Lu et al., 2025 [52]
	Aquatic biological resources	Reduce fishing	He and Chen, 2022 [53]
Optimizing production methods	Farmlands	Reducing chemical inputs Appling organic fertilizer Planting green manure Crop rotation comprehensive utilization of crop straw	Liu et al., 2020 [54] Yi et al., 2021 [55] Ntakirutimana et al., 2019 [56] Zhang et al., 2025 [57] Jiang et al., 2025 [58]
	Grasslands	Manual grass-cutting Breed improvement Cross-regional grazing Barn feeding	Li and Liu., 2023 [21]
	Forests	Agroforestry Cultivating economic forests	Zhu and Wang, 2015 [59] Liu et al., 2025 [60]
	Watersheds	Paddy land to dry land	Hong et al., 2020 [61]
Changing livelihood types	Multi-field	Non-farm livelihood	Wen et al., 2022 [62] Xu and Feng, 2025 [63]

4. The Influencing Factors of Agricultural Households' Response to Eco-Compensation in China

The enthusiasm of agricultural households in responding to policies directly affects the efficiency of compensation. This also makes its influencing factors one of the most prominent research hotspots in the field of eco-compensation in China. At present, academic research on the influencing factors of agricultural households' response to eco-compensation generally includes two perspectives: The first is the influencing factors of response willingness, analyzing which factors affect agricultural households' psychological

acceptability [64]. The second is the response result, analyzing which factors drive the actual behaviors of agricultural households [65]. These two perspectives have their own advantages and disadvantages. The analysis of willingness helps eliminate interference from other contexts, allowing research to focus on agricultural households' recognition and expectations of eco-compensation. But there are often varying degrees of differences between the willingness and the actual outcomes. The analysis of actual behavior focuses on factual results and has higher reliability. But some actual behaviors are formed by various contexts for a long time. It is difficult to clarify the relationship between the behaviors and eco-compensation in research. However, whether from the perspective of intention or outcome, this study classified the influencing factors into three categories, namely policy mechanism (PM), objective conditions (OCs), and subjective attitude (SA) (Figure 2).

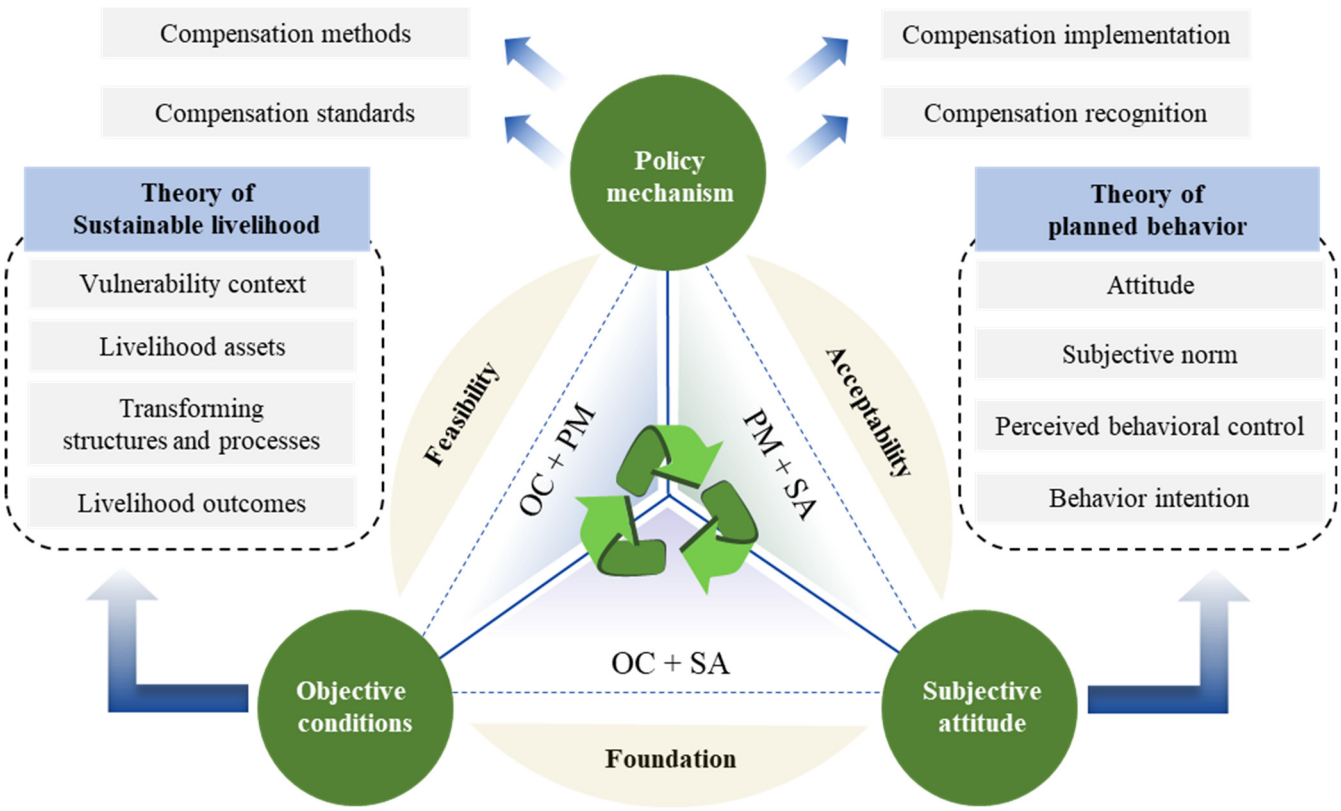

Figure 2. Schematic diagram of the types of influencing factors for agricultural households' responses.

The first is a policy mechanism. It is a combination of eco-compensation rules and implementation. The core rule of ecological compensation in China is the methods and standards [19]. The compensation methods generally include two types: transfusion type and hematopoietic [66], with specific paths including providing subsidies, training, and materials [67]. There are differences in agricultural households' preferences for various methods [65,68]. The compensation standard represents the intensity of compensation in a specific method, typically expressed as the amount of compensation, and it also influences the livelihood choices of agricultural households [69]. In addition, the implementation of policies represented by supervision and contract duration [70,71], as well as the satisfaction and trust of agricultural households towards the entire compensation mechanism, is also an important factor that affects their responses [21,22]. Overall, the impact of the policy mechanism on agricultural households' responses stems from their preferences and experiences regarding the rules and their implementation.

Under the influence of the policy mechanism, the final response of agricultural households is also constrained by their own factors, which include both objective factors and

subjective factors. The specific indicators of these two types of factors can generally be covered by two theories, namely the theory of sustainable livelihoods and the theory of planned behavior (Table 2). And these two theories were also frequently mentioned in the literature and correspond to different research perspectives [21,72].

When analyzing objective factors, the theory of sustainable livelihoods is applicable. The relevant indicators reflect the capital and capabilities of agricultural households in responding to eco-compensation. This study, based on the analysis framework of sustainable livelihood theory, sorted out the relevant indicators that appeared in the literature. These indicators belong to vulnerability contexts [73,74], livelihood assets [71,75–78], transforming structures and processes [79], and livelihood outcomes [21]. Among them, indicators related to livelihood assets appeared frequently in previous studies. Livelihood assets reflect the resources and capabilities of agricultural households to resist shocks and transform their livelihood strategies [80,81]. The sustainable livelihood theory divides livelihood assets into five categories: human capital, social capital, natural capital, material capital, and financial capital. Some studies may also expand into information capital and cultural capital [81,82]. It is worth noting that the influencing factor indicator system constructed based on sustainable livelihood theory was often dominated by objective indicators.

When analyzing subjective factors, the theory of planned behavior provides an applicable framework. The relevant indicators capture agricultural households' cognition and intentions regarding eco-compensation responses. The theory of planned behavior is a core theory in social psychology research. This theory suggests that behavioral intention is related to behavior, and behavioral intention can be predicted through attitudes toward the behavior, subjective norms, and perceived behavioral control [83]. In this type of research, scholars mostly construct indicator systems based on the above factor types and then analyze the factors that affect agricultural households' responses through statistical methods.

In addition, in some empirical studies, scholars combined different types of factors to construct analytical frameworks for a comprehensive analysis of influencing factors. For example, research that combined sustainable livelihood theory with compensation mechanisms not only included the impact of policies but also reflected the conditions for agricultural households to respond, which belongs to a comprehensive analysis of objective factors [21]; research that combined the theory of planned behavior with compensation mechanisms not only reflected the subjective willingness of farmers to respond but also demonstrated their views of policy mechanisms, which belongs a comprehensive analysis of subjective factors [72]. Of course, subjectivity and objectivity are not completely separate, which usually manifests in two aspects. Firstly, some studies have integrated the theory of planned behavior and sustainable livelihood theory to establish econometric models, achieving a combination of subjectivity and objectivity [84]. Secondly, there are studies that independently construct a factor system that simultaneously includes both subjective and objective indicators [65].

Table 2. Summary of factors influencing agricultural households' responses.

Factor Types	Subclasses	Specific Indicators or Factor Descriptions	References
Compensation policy and mechanism	Compensation methods	Fund compensation Technology compensation	Yang and Zheng, 2021 [65]
	Compensation standards	Subsidy amounts	Qiu et al., 2022 [69]
	Compensation implementation	Supervision intensity Contract duration	Pang and Jin, 2021 [70] Giefer et al., 2021 [71]
	Compensation recognition	Satisfaction Level of trust in policy	Li and Liu., 2023 [21] Xie et al., 2017 [22]

Table 2. *Cont.*

Factor Types	Subclasses	Specific Indicators or Factor Descriptions	References
Theory of sustainable livelihood	Vulnerability contexts	Location characteristic Road infrastructure Industrial structure Economic growth	Démurger and Pelletier, 2015 [73] Liu et al., 2024 [74]
	Livelihood assets	Household size or number of agricultural workers Education level Farmland size Farmland quality Household income Feasibility of loan Possession of houses Possession of livestock Possession of agricultural machines Possession of durable consumer goods Social network Energy structure	Giefer et al., 2021 [71] Sheng et al., 2019 [75] Wei et al., 2021 [76] Liu et al., 2018 [77] Shen et al., 2025 [78]
	Transforming structures and processes	In addition to eco-compensation, it also includes related supporting systems such as social security and employment assistance.	Chen, 2023 [79]
	Livelihood outcomes	Income change Food safety change Disaster resilience change	Li and Liu., 2023 [21]
Theory of planned behavior	Attitudes toward the behavior	Positive/negative attitudes towards protecting/destroying the ecology Perception of ecological and economic benefits of compensation policies	Deng et al., 2016 [85] Peng et al., 2022 [64]
	Subjective norms	Pressure from neighbors Trust in the government and people around	Shi et al., 2019 [86] Dai et al., 2025 [84]
	Perceived behavioral control	Capacity Policy perception	Shi et al., 2019 [86] Jin et al., 2020 [72]
	Behavioral intention	The willingness of households to respond	Pang et al., 2021 [87]

5. The Effects of Agricultural Households' Response to Eco-Compensation in China

As mentioned earlier, China's eco-compensation has dual economic and ecological goals. Therefore, evaluating the effects of agricultural households' responses should also be considered from both economic and ecological perspectives (Table 3). Firstly, in terms of the economy, scholars were concerned about the impact of eco-compensation policies on agricultural households' income. Generally speaking, eco-compensation changes agricultural households' income through two key points. The first is compensation funds provided by the government to agricultural households, which not only provide a stable source of income for agricultural households but also change their income structure [52,88]. The second is that after agricultural households optimize their production methods or expand their livelihood types, their economic benefits may change accordingly [89]. As research deepens, scholars' focus has shifted from income levels to issues such as income inequality [90], cost–benefit ratio [91], livelihood asset index [92], and energy transition [93]. Overall, agricultural households' response to eco-compensation not only helps improve their income level and optimize their income structure but also promotes social equity to a certain extent and drives the development of the industry.

As for ecological effects, unlike economic effects, it is difficult to conduct micro-level evaluations at the household scale. Scholars used to assess a certain area through remote sensing technology or monitoring data [94,95]. The strategy followed in this evaluation is often to compare the relevant ecological indicator values before compensation (agricultural households have not responded) and after compensation (agricultural households have responded) in the same region [91,96]. The advantage of this strategy is that it can directly quantify the ecological changes before and after compensation. But the disadvantage is that this change may be caused by multiple factors, making it difficult to separate the impact

of factors other than compensation on the results. But overall, research showed that the ecological effects in various fields are often positive [91,94–96].

Table 3. Summary of typical studies on the ecological and economic effects of compensation.

Effect Types	Study Areas	Evaluation Objects	Key Conclusions	References
Economy	Wolong nature reserve in Sichuan	Income growth	Increased PES participation can promote income growth. And PES participation was more beneficial to small and medium farmers than large farmers.	Sheng and Wang, 2022 [88]
	Miyun reservoir in Beijing and Hebei	Total household income	The paddy land-to-dry land program had a positive but somewhat limited effect on total household income (increase of 2171 CNY/person).	Li et al., 2018 [89]
	Nationwide (meta-analysis)	Income with different sources	The Grain for Green Program had a significant positive impact on total income (including subsidies) and off-farm income.	Lu et al., 2025 [52]
	Zhouzhi county, in Shaanxi	Income inequality	Overall, income inequality was less among households participating in the Sloping Land Conversion Program than among those that did not after 7 y of the PES program.	Li et al., 2011 [90]
	Miyun reservoir in Beijing and Hebei	Benefit–cost ratio	The paddy land-to-dry land program has an overall benefit–cost ratio of 1.5, and both downstream beneficiaries and upstream providers gain from the program.	Zheng et al., 2013 [91]
	The middle and lower reaches of the Yangtze River	Livelihood assets	The total index of livelihood assets of retired fishery households has significantly increased, and the coupling coordination degree index of livelihood assets of retired fishery households has significantly increased.	He et al., 2023 [92]
	Black river basin in Shaanxi	Energy transition	The Grain-to-Green Program promoted energy transition both directly and indirectly by increasing non-farm income.	Wang et al., 2025 [93]
Ecology	Nationwide	Grassland ecology	NDVI increased by 1.2% in grassland eco-compensation regions from the pre-program period (2006–2010) to the program period (2011–2015).	Hou et al., 2021 [94]
	Checheng and Jichang in Shanxi	Forest ecology	During 2002–2014, the proportion of the study area in forest cover increased from 57.9% to 63.2%.	Zhang et al.,2020 [95]
	Heilonggang low plain area in Hebei	Farmand ecology	In the past two years since the implementation of the crop rotation and fallow system, the soil organic matter in the demonstration zone has increased by 8.6%, and the average content of total nitrogen, available phosphorus, and available potassium in the soil has increased by 15.8%.	Zhao et al., 2018 [96]
	Miyun reservoir in Beijing and Hebei	Watershed ecology	The paddy land-to-dry land program increased water yield by 1.82×10^7 m^3 per y and reduced total nitrogen and total phosphorus by 10.36 and 4.34 tons per y, respectively. And the increase in water yield is 5% of the average runoff in Miyun Reservoir between 2000 and 2009.	Zheng et al., 2013 [91]

Although these ecological and economic effects are generally positive, some controversies are worthy of attention. The first is the balance between production and ecology. The limitations imposed by eco-compensation on resource utilization may lead to difficulties for related industries in maintaining raw material supply [97]. Research should not focus solely on the ecological effects while neglecting the necessary agricultural production. The balance between protection and development deserves attention. The second is the differences in effects among individuals. The overall positive effect does not mean that there are no differences among individuals [98]. For those agricultural households that benefited little from the compensation, differentiated policy measures are needed to motivate them.

This is conducive to promoting social equity and enhancing the enthusiasm of agricultural households to respond. The third is the other externalities caused by eco-compensation. For example, studies have shown that although the paddy land-to-dry land program implemented in the Miyun Reservoir increased water yield, it also increased the application rates of nitrogen and phosphorus [91]. This might offset some of the ecological effects. The fourth is the challenge of achieving sustainable effects. Once compensation stops or funds decrease, agricultural households may resume their improper use of resources. To prevent this situation, efforts should be made to promote technological eco-compensation models and support agricultural households in establishing stable ecological industries.

6. Discussion

This review presented China's eco-compensation experience to scholars worldwide, particularly focusing on the incentives for agricultural households' responses and their effects. Although most of the cases are from China, many of its research conclusions have reference value for the PES worldwide.

In terms of theory, the research framework presented in this review is also applicable to PES. In PES and AES research, the participation of agricultural households and the final ecological and economic outcomes also hold a significant position [99,100]. The multi-response of this study could provide new ideas for PES research. Regarding the influencing factors of agricultural households' response/participation, the research on agricultural households' participation in the PES or AES also focuses on mechanism design, as well as the subjective attitudes and objective conditions of agricultural households [101–103]. The indicator system of the influencing factors derived from this study involves the theory of sustainable livelihood and the theory of planned behavior, both of which are universal academic theories. They are applicable in general PES or AES scenarios [104]. As for the effects, this study involved macroscopic ecological effects, which could be used as a reference for large-scale PES projects.

In terms of practice, China's eco-compensation is a large-scale public institutional arrangement, emphasizing government leadership. While PES places greater emphasis on market-based approaches, the apparent gap between the two frameworks is narrowing in practice. In China, the market mechanism for eco-compensation is developing rapidly, and it has achieved success, particularly in watershed horizontal compensation [105,106]. Furthermore, new market mechanisms such as forest ecological banks and tourism eco-compensation possess the same flexibility as small-scale PES [107,108]. At the same time, due to the advantages in fundraising, management, and promotion, the government also plays an important role in PES and AES [39,46]. Therefore, the market-oriented development of the eco-compensation and the significance of government support for PES have made the connection between the two increasingly close. During the compensation/payment process, particularly regarding the motivation of agricultural households, the experiences from both approaches offer valuable lessons for mutual learning.

In the context of China's rural revitalization agenda, large-scale eco-compensation serves as a driver of economic development. Consequently, scholars have paid particular attention to changes in agricultural households' income and livelihood strategies [109,110]. For PES and AES programs, achieving economic benefits may be a less urgent priority compared to eco-compensation schemes. Nevertheless, numerous studies have examined their economic effects [99,111]. This might be carried out to achieve multiple goals or to promote the continuous participation of agricultural households.

7. Conclusions

This article reviewed the connotation, influencing factors, and ultimate effects of agricultural households' response to eco-compensation in China. The results indicated that agricultural households' response to eco-compensation mainly includes reducing production intensity, optimizing production methods, and changing livelihood types. These three types of responses are not mutually exclusive. On this basis, taking protected objects such as cultivated lands, grasslands, forests, watersheds, and aquatic biological resources as examples, the specific connotation of agricultural households' response was explained. Subsequently, according to policy mechanisms, sustainable livelihood theory, and planned behavior theory, the factors that affect agricultural households' responses have been sorted out, forming a systematic factor indicator system framework. Finally, focusing on the policy objectives of eco-compensation, the research progress on the economic and ecological effects of agricultural households' responses and the final results were summarized.

Although current research on eco-compensation in China is increasingly emphasizing the response of agricultural households, there are still some shortcomings specifically manifested in the following four aspects.

Firstly, there is an imbalance in research fields. China's *Regulations on Ecological Protection Compensation* cover forests, grasslands, wetlands, deserts, oceans, watersheds, and farmland, as well as other important ecological and environmental elements such as aquatic biological resources, terrestrial wildlife, and plant resources stipulated by laws, administrative regulations, and national stipulations. Obviously, there is a lack of research on agricultural households' responses in areas such as deserts, oceans, and terrestrial wildlife resources. Meanwhile, the proportion of research on the responses of agricultural households in important protected areas represented by national parks is relatively small.

Secondly, the impact of context is not comprehensive. Sustainable livelihoods is one of the most fundamental theories for studying agricultural households' responses. However, current research often focuses more on the impact of livelihood assets on responses, and the understanding of the impact from vulnerability contexts is not comprehensive enough. Factors such as policies, markets, and culture have a significant impact on agricultural households' livelihood decisions, but currently, research tends to focus on surveys within households, with limited understanding of external contexts.

Thirdly, the ecological effects are not specific. As mentioned earlier, current research on ecological effects is mostly conducted from a macroperspective. This perspective weakens the role of agricultural households' responses, which is reflected in two aspects: The first issue is that the effects of natural factors are difficult to separate, resulting in the inability to accurately extract the compensation effects. The second is that the effectiveness of various responses from agricultural households is integrated together, making it difficult to distinguish the roles of each response.

Fourthly, the strategies for improvement are not detailed enough. Existing agricultural household response studies have proposed related strategies for improving mechanisms. But these strategies are always limited to simple policy recommendations and ideas. Many compensation improvement mechanisms still require scientific verification; otherwise, they cannot support policy improvements.

8. Future Directions

Based on the above shortcomings, this study proposes the following suggestions for future research (Figure 3).

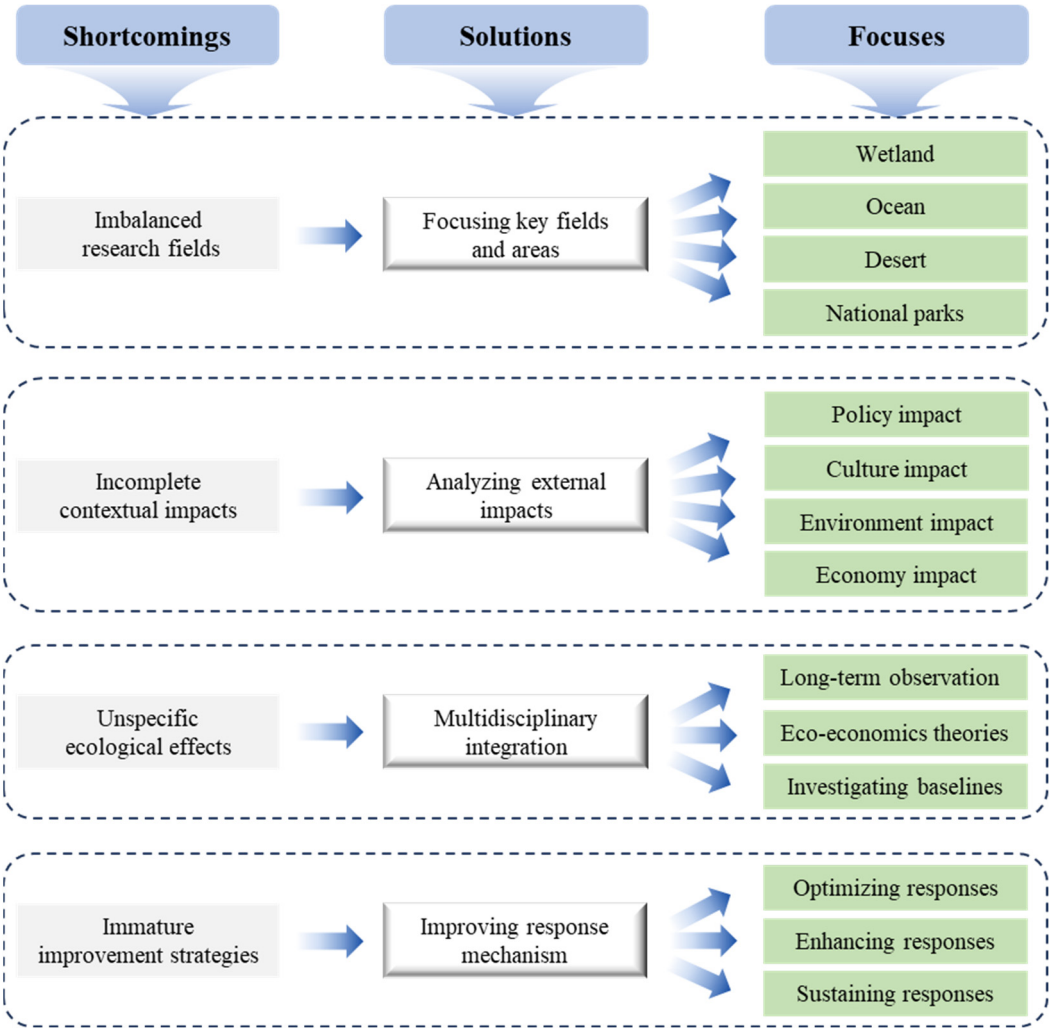

Figure 3. Future research suggestions on eco-compensation.

1. Strengthening research in key areas and protected areas

Systematic research on compensation mechanisms and agricultural household responses in fields such as wetland, ocean, and desert should be given due attention in the future. In addition, China's *Regulations on Ecological Protection Compensation* not only clarify the classification of eco-compensation fields but also propose to establish an ecological protection compensation mechanism for natural protected areas with national parks as the main body. National parks are important protected objects with extremely high ecological value worldwide. Guiding agricultural households to scientifically respond to eco-compensation is one of the important paths to coordinate the protection and development of national parks. It is necessary to focus on the collaborative strategy between agricultural households' responses and national park protection goals.

2. Focusing on the correlation between contexts and agricultural households' responses

Many external context factors deserve attention, including at least the following four points: The first is policy synergy. Against the backdrop of China's vigorous promotion of rural revitalization, in addition to eco-compensation, there are many policies and subsidies that benefit agriculture. They may all have certain impacts on the production and life of agricultural households. Future research could focus on how to improve the efficiency of policy resource utilization by promoting policy synergy. The second is the influence of culture. China has a rich traditional culture. Many cultural customs contain the connotation

of harmonious coexistence between humans and nature. How to use the value of these cultures to empower agricultural ecological transformation is worth considering. The third is the impact on the natural environment. The impact of the natural environment on agricultural households' livelihoods is evident, and in some ecologically fragile areas, the impact of factors such as climate change and natural disasters on their response should be given priority consideration. The fourth is the development of the economy. The level of regional economic development may be one of the important factors affecting the response of agricultural households in different regions of the same ecosystem, and this impact could become an important basis for formulating differentiated policies.

3. Multidisciplinary integration for precise evaluation of ecological effects

As mentioned earlier, eco-compensation is the process of indirectly restoring ecological functions by incentivizing agricultural households to respond through economic means. Therefore, from the perspective of policy mechanisms, the ecological effects of compensation belong to the interdisciplinary field of ecology and economics. Only through the integration of disciplines can more accurate evaluations be made. The following aspects could be referred to: Firstly, conduct micro-level ecological effect research through long-term positioning observation and combine digital management technology to analyze the effects of agricultural households' responses in nearby areas using the control variable method, which can help eliminate the influence of natural factors. Secondly, quantify the economic value of ecological effects through ecological economics theory. This brings the possibility of calculating the cost-benefit of compensation. The results not only contribute to the establishment of market mechanisms but also provide decision-makers with feedback on compensation outcomes. Thirdly, strengthen ecological baseline surveys. Fully consider the marginal benefits of ecological restoration and set appropriate ecological protection goals based on different ecological conditions and background situations.

4. Studying the improvement mechanism of agricultural households' responses

After discovering numerous factors that affect agricultural households' responses, we believe that future research on mechanism improvement can be carried out in the following three aspects: The first is how to optimize the response. Since agricultural households' responses are diverse, it is necessary to study how to identify and guide them to make the optimal response that is suitable for their current family situation and meets their compensation needs. This is of great significance for improving compensation efficiency and ensuring the livelihood security of agricultural households. The second concerns how to enhance household responses. Based on the analysis of influencing factors, study the path to improve agricultural households' response and conduct a feasibility analysis. How to overcome the long-term behavioral inertia of agricultural households and build an efficient supervision and management system deserves special attention. The third is how to continuously respond. In cross-sectional studies, it is difficult to identify interruptions in agricultural households' responses. Long-term tracking research could be conducted on agricultural households to identify the key reasons for the interruption of their responses and explore long-term mechanisms to support sustained responses.

Finally, from a global perspective, this study suggests that future work should focus on experience sharing and efficiency improvement. For instance, effective government management is one of the distinctive features of China's eco-compensation. This is of great help in achieving the multiple benefits of compensation, especially in income growth. However, the efficiency of the compensation might be relatively low. PES places greater emphasis on market-oriented approaches. This helps enhance the accuracy and efficiency. But it also poses challenges for large-scale activities and the stability of funds. Whether it is eco-compensation or PES, how to integrate the advantages of both and better leverage the

roles of the government and the market is what future research needs to explore. This is of vital importance for improving the response and achieving the final benefits.

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